

LOHITHA BANDARU

Senior AI/ML Architect · Generative AI · MLOps · Enterprise Pricing Intelligence
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Executive profile

Senior AI/ML Architect with 16+ years translating ambiguous commercial problems into production intelligence. Recognized for architecting enterprise pricing platforms that pair Generative AI, agentic orchestration, and rigorous MLOps with the operational discipline of SAP S/4HANA and PriceFx. Trusted partner to finance and commercial leadership in quantifying margin lift, accelerating quote velocity, and replacing static rule engines with adaptive, explainable AI. Bring an executive lens to every build: outcome first, governance always, value measured in basis points.

Signature outcomes

- **Margin Expansion at Scale.** Architected an AI pricing platform serving a multi-billion-dollar catalog, lifting realized margin through elasticity-driven recommendations validated via A/B and shadow deployments before any production promotion.
- **Quote Velocity.** Delivered sub-200 ms inference for real-time quote scoring during peak commercial flows, compressing quote-to-cash cycles and unlocking front-line seller productivity.
- **Generative AI in Production.** Stood up a RAG assistant grounded in customer agreements, historical pricing rationale, and CPL governance, giving commercial teams instant, explainable context inside their workflow.
- **Governed ML Operations.** Instituted Vertex AI Model Registry, lineage capture, drift monitoring, and versioned artifacts, turning model deployment into an auditable, board-defensible discipline.
- **Executive Alignment.** Quantified uplift in finance terms and presented model ROI through quarterly executive dashboards, securing repeated reinvestment in the AI roadmap.

Core capabilities

AI & Generative Systems

LangGraph agentic orchestration, LangChain, Retrieval-Augmented Generation, Vertex AI, vector embeddings (pgvector), price elasticity and discount uplift modeling, responsible AI guardrails

Cloud & Data Platforms

Google Cloud (AlloyDB, BigQuery, Vertex AI, Cloud Run, Cloud Functions, Pub/Sub, Dataflow, Cloud Composer), Azure, SAP S/4HANA, PriceFx SaaS

MLOps & Engineering Discipline

Airflow, MLflow, model registry and versioning, drift monitoring, A/B and shadow deployments, CI/CD with Jenkins, Docker, Kubernetes

Pricing & Commercial Systems

PriceFx native modeling, CPQ and quote-to-cash, list price and promotion strategy, CPL governance, dynamic discounting

Languages & Frameworks

Python, Java 8/11/17, Groovy, Spring Boot, FastAPI, REST and GraphQL, JAX-RS, microservices

Data Engineering

AlloyDB, PostgreSQL, Oracle, DB2, Microsoft SQL Server, MongoDB, Redis, stored procedures, performance tuning, SQL optimization

Leadership Practice

Target-state architecture, executive storytelling, mentorship across data, MLOps and application teams, agile delivery, code review culture

Professional experience

Mohawk Industries Inc. · Calhoun, GA Sep 2022 – Present

Senior AI/ML & Pricing Architect · AI-Driven Pricing Intelligence on GCP, SAP S/4HANA, PriceFx

Own end-to-end architecture and delivery of an AI-powered pricing platform replacing legacy rule-only engines with adaptive, explainable recommendations across list price, promotion, discount, and quote workflows for a multi-billion-dollar product catalog.

- **Target Architecture.** Defined the GCP target state with AlloyDB as the operational analytics and pgvector embedding store, enabling high-throughput feature retrieval across millions of customer-product combinations.
- **Agentic Orchestration.** Built a LangGraph agentic layer sequencing list price calculation, promotion eligibility, discount optimization, and quote shaping across heterogeneous data sources and downstream systems.
- **Real-Time Inference.** Deployed inference on Cloud Functions and Cloud Run with autoscaling, sustaining sub-200 ms response for quote scoring during peak commercial flows.
- **MLOps at Scale.** Engineered Airflow (Cloud Composer) pipelines for feature engineering, scheduled retraining, drift detection, and lineage capture across elasticity and uplift models.
- **Generative AI Assistant.** Delivered a RAG assistant surfacing historical pricing rationale, customer agreements, and CPL context to commercial teams, grounded in AlloyDB-backed embeddings.
- **Model Governance.** Established Vertex AI Model Registry, versioned artifacts, A/B experiments, and shadow deployments, validating margin lift before any production promotion.
- **PriceFx Extensibility.** Authored Groovy logic extensions that invoke ML inference inside PriceFx, injecting model outputs directly into list price, promotion, and quote pipelines with full audit trail.
- **Data Contracts & Observability.** Stood up Pub/Sub event contracts and lineage instrumentation, ensuring traceable flow from raw transactional data to features to model decisions.
- **Executive Storytelling.** Partnered with finance and commercial leadership to quantify uplift and present model ROI in quarterly executive dashboards, securing continued AI investment.
- **Talent Multiplier.** Mentored engineers across Python, Airflow DAG patterns, prompt engineering, and responsible AI guardrails, building a durable internal AI bench.

Intone Networks · Client: State of Georgia, Department of Labor Sep 2021 – Aug 2022

Senior Lead Developer · Pandemic Unemployment Assistance & Virtual Claims Agent

- **Mission-Critical Delivery.** Delivered a high-availability claims platform serving self-employed workers, freelancers, and independent contractors, processing intake, eligibility, and benefit disbursement at state scale during peak crisis demand.
- **Service Architecture.** Designed Java and Spring Boot microservices fronted by REST APIs aligned to OpenAPI 2.0, integrated with relational and NoSQL stores for audit-grade reporting.
- **Conversational AI Pilot.** Prototyped an early virtual claims agent for claim-status inquiries, materially reducing call-center load during peak periods.
- **Release Governance.** Owned change management across SIT, UAT, DR, and Production, including load-balancer cutover and rehearsed rollback playbooks.

Intone Networks · Client: Amway Mar 2017 – Aug 2021

Senior Software Developer · MAGIC – Global Distributor Management

- **Global Platform Architecture.** Architected REST and SOAP services for distributor registration, contracts, renewals, reactivations, and lifecycle events across Amway's global markets.
- **Single Customer View.** Designed DB2 stored procedures and tuned SQL access paths to unify identity, address, communications, privacy, tax, and related-party data into a single customer view.
- **Throughput Engineering.** Owned IBM WebSphere sizing, JVM tuning, and IO partitioning to sustain steady throughput under heavy distributor activity and built multi-threaded synchronization workers keeping load-balanced caches consistent across data centers.
- **Engineering Leadership.** Led code reviews and mentored offshore engineers, driving enterprise change management with rollback strategies, upstream and downstream impact notifications, and post-deploy validation.

GOMIAPP · Client: Virtusa (Dell SecureWorks) Sep 2016 – Nov 2016

Senior Software Engineer · Mobile Security Dashboard

- **Incident Modernization.** Built a mobile incident-escalation platform pushing security alerts directly to end users, replacing manual call-out workflows for the SecureWorks tech support team.
- **Service Engineering.** Implemented RESTful services with HATEOAS, JMS-MQ async messaging, Spring AOP audit, and MyBatis hierarchical MySQL persistence; automated CI with Maven and Jenkins.

GOMIAPP · Client: Anixter Inc. Apr 2016 – Aug 2016

Senior Full-Stack Developer · Customer Program Profile Pricing

- **Real-Time Pricing Intake.** Delivered a global pricing intake web application that fed real-time data into the PROS pricing engine for Anixter's manufactured-goods catalog.
- **Full-Stack Engineering.** Built PrimeFaces and JSF MVC front end with REST and SOAP integrations, Spring JdbcTemplate persistence, and Spring AOP auditing.
- **Performance Discipline.** Authored DB2 stored procedures for hierarchical pricing and bulk loads; monitored JVM health with Wily and tuned thread schedulers across load-balanced servers.

JPMorgan Chase · Hyderabad, India Sep 2012 – Feb 2016

Senior Application Developer · IVR Configuration & Emergency Rules Middleware

- **Always-On Routing.** Owned middleware managing emergency call routing, business-hours automation, and configuration across heterogeneous IVR systems built on .NET, EJB, SOAP, REST, and VXML.
- **Service Design.** Engineered services using PrimeFaces and JSF, Spring WebFlow, REST and SOAP, JMS-MQ, and JPA / Hibernate; provided Level 3 production support and led post-production stabilization.
- **Release Automation.** Engineered automated release scripts with SSH for promotions across environments and led code reviews using ClearCase version control.

Infosys · Client: Bank of America · Chennai, India Apr 2011 – Apr 2012

Technology Analyst · Informatica Migration Automation

- **Process Automation.** Built a web-based migration tool automating Informatica code promotions across 7 environments and 140 applications with multi-level architectural approvals and scheduled migration triggers.
- **Audit Readiness.** Reduced manual effort during external and internal audits and supported migration tracking and stakeholder communications.

Satyam Technologies · Client: Scotiabank · India / Canada Dec 2007 – Feb 2010

Onsite Coordinator & Developer · Multi-Factor Authentication

- **Security Modernization.** Integrated Entrust Identity Guard into Scotia Online, introducing machine-based authentication and challenge-question flows on top of password login.
- **Onshore / Offshore Delivery.** Coordinated with offshore teams, gathered requirements directly with the client, and led deployment across SIT, UAT, and Production.

American Express · Chennai, India Oct 2005 – Sep 2007

Software Engineer · File Services – Global Tracking & Monitoring

- **Operational Visibility.** Built a global web console giving business and technology teams real-time visibility into file transfer, validation, and transformation between American Express and external submitters.

Education

Bachelor of Computer Science · Hyderabad, India 2005

References

Available upon request.